INCH-POUND

MIL-DTL-55036B
12 October 2005
SUPERSEDING
MIL-C-55036A(EL)
9 May 1973

DETAIL SPECIFICATION

CABLE, TELEPHONE, WM – 130( )/G

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

Inactive for new design after 14 September 1999

1. SCOPE

1.1 Scope. This specification covers the requirements of a 26 pair telephone cable, insulated with polyethylene, armored with a galvanized steel braid, and covered with a polyvinylchloride low temperature jacketing compound. This cable is designated as Cable, Telephone WM-130 ( )/G (see 6.8).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAI, P.O. Box 3990, Columbus, OH 43218-3990, or email to WireCable@dscc.dla.mil. Since contact information can change, please verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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FEDERAL SPECIFICATIONS

L-P-390	-	Plastic, Molding and Extrusion Material, Polyethelyne and Copolymers, (Low, Medium and High Density).

COMMERCIAL ITEM DESCRIPTIONS

A-A-59551	-	Wire, Electrical, Copper, (Uninsulated)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-631	-	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-DTL-12000	-	Cable, Cord, and Wire, Electric; Packing of

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CECOM DWG SM-B-159154	-	Grip Cable

(Copies of these documents are available online at http://www.monmouth.army.mil/CELCMS/lrc/lrc.html or from the U.S. Army Communications and Electronics Materiel Readiness Command Logistics Engineering Director, 12 WD Bldg 601 McAfee Center, Fort Monmouth, NJ 07703.)

2.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM-A90/A90M	-	Standard Test Method for Weight [Mass] of Coating on Iron and Steel Articles with Zinc or Zinc - Alloy Coatings.
ASTM A239	-	Standard Practice for Locating the Thinnest Spot in a Zinc (Galvanized) Coating on Iron or Steel Articles.
ASTM D573	-	Standard Test Method for Rubber Deterioration in an Air Oven.
ASTM D1248	-	Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable.

(Copies of these documents are available online at http://www.astm.org or from ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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2.4 Order of precedence. In event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Cable construction. The cable shall be constructed in accordance with the following requirements:

3.1.1 Conductor. The conductors shall conform to type C, class B, of A-A-59551, except that the conductors shall consist of six coated copper strands concentrically stranded around one galvanized steel strand (see 4.6.1).

3.1.1.1 Copper strands. Each coated copper strand shall be 32 American Wire Gauge (AWG) soft or drawn and annealed wire in accordance with type S of A-A-59551 (see 4.6.1.1).

3.1.1.1.1 Galvanized steel strand. The steel strand shall have a diameter of 0.008 inch (0.20mm) ± 5% and shall be covered with a coating of commercial pure zinc (galvanized). The strands shall meet the following requirements: (see 4.6.1.2).

a. Tensile strength. Each strand shall have a minimum breaking strength of 300,000 pounds per square inch (see 4.6.1.2a).

b. Weight of coating. The weight of the zinc coating shall be not less than 25 grams per kilogram of coated strand (see 4.6.1.2b).

c. Uniformity of coating. The zinc coating shall be smooth, continuous, free of pinholes and splits, and shall adhere firmly to the steel wire. It shall withstand 1 or more immersions before the end point is reached when tested as required in 4.6.1.2c, but the number of immersions shall be approximately the same for all specimens from the same sample.

3.1.2. Conductor insulation. Each conductor shall be insulated with 0.010 inch (.0254 mm) of high-density polyethylene having a nominal wall thickness of 0.010 inch (.0254 mm). The minimum wall thickness shall be not less than 0.008 inch (0.20 mm). The insulation shall be smooth, and free from holes and occlusions. The polyethylene shall be in accordance with L-P-390, type II, class H, grade 1 except that the density shall be .941 - .965 inch (23.90 - 24.51 mm) and melt index shall be 0.50 inch (12.7 mm) maximum. Pigments used to color the insulation shall be compatible with the insulation antioxidant (see 4.6.2).

3.1.2.1 Thermal stress cracking. Six specimens of insulated conductor removed from a finished cable shall be subjected to the test specified in 4.6.2.1. Following the test there shall be no evidence of cracking of the insulation when examined under a magnification of 3 diameters (focal distance 8 cm).

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3.1.3 Color code. The conductor shall be fully color-coded. The different colors used shall conform to table I and shall be readily distinguished. The base color shall be an integral part of the insulation. The color band shall be applied using nonconductive ink over the insulation with a minimum circumferential coverage of 80 percent. The width of the band shall be approximately .125 inch (3.175 mm). Spacing between the 2 bands shall be approximately 0.0625 inch (1.586 mm). The color coding of the wire shall repeat every .0.75 inch (19.05 mm) (see 4.6.3).

TABLE I. Color code.

Pair number	Layer	Wire base color	Wire band	Mate base color	Mate band
1	Center	Blue	1 White	Blue	2 White
2	Center	Orange	1 White	Orange	2 White
3	Center	Green	1 White	Green	2 White
4	First	Brown	1 White	Brown	2 White
5	First	Gray (Slate)	1 White	Gray (Slate)	2 White
6	First	Blue	1 Red	Blue	2 Red
7	First	Orange	1 Red	Orange	2 Red
8	First	Green	1 Red	Green	2 Red
9	First	Brown	1 Red	Brown	2 Red
10	First	Gray (Slate)	1 Red	Gray (Slate)	2 Red
11	First	Blue	1 Black	Blue	2 Black
12	First	Orange	1 Black	Orange	2 Black
13	Second	Green	1 Black	Green	2 Black
14	Second	Brown	1 Black	Brown	2 Black
15	Second	Gray (Slate)	1 Black	Gray (Slate)	2 Black
16	Second	Blue	1 Yellow	Blue	2 Yellow
17	Second	Orange	1 Yellow	Orange	2 Yellow
18	Second	Green	1 Yellow	Green	2 Yellow
19	Second	Brown	1 Yellow	Brown	2 Yellow
20	Second	Gray (Slate)	1 Yellow	Gray (Slate)	2 Yellow
21	Second	Blue	1 Violet	Blue	2 Violet
22	Second	Orange	1 Violet	Orange	2 Violet
23	Second	Green	1 Violet	Green	2 Violet
24	Second	Brown	1 Violet	Brown	2 Violet
25	Second	Gray (Slate)	1 Violet	Gray (Slate)	2 Violet
26	Second	Red	1 White	Red	2 White

3.1.4 Pair twist. The insulated conductors shall be twisted into pairs. The average length of lay in any pair of finished cable shall not be less than one inch or more than six inches when measured in any ten-foot length. Each pair should have a varied lay (see 4.6.4).

3.1.5 Cabling. The twisted pairs shall be arranged in layers, 1 pair thick, to form a cylindrical core. The center layer shall consist of 3 pairs: the 1st layer 9 pairs, and the 2nd layer 14 pairs shall be cabled with a left-hand lay of not more than 5.0 inches (127 mm) (see 4.6.5).

3.1.6. Marker. Manufacturer's markers are permitted. The use of markers shall have no deleterious effect on the finished cable (see 4.6.6).

3.1.7 Binder. An open binder of nylon yarn may be applied over the core with a right-hand lay, if necessary, in the production process. (see 4.6.7).

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3.1.8 Core wrap. The core shall be covered with an approved 0.001 inch (.0254 mm) polyethylene terephtalate tape, type G, forms F or T_f, class I, in accordance with MIL-I- 631. The tape shall be applied helically with an overlap to provide at least 2 thicknesses of material over the core (see 4.6.8).

3.1.9 Inner jacket. An inner jacket comprised of a tight fitting polyethylene compound, type I, class A, category 5, grade E4 or E5 as specified in ASTM D1248 shall be applied over the core wrap. Minimum wall thickness shall be not less than 0.014 inch (.36 mm) (see 4.6.9).

3.1.10 Braid. A galvanized steel braid shall be applied over the inner jacket. The braid shall consist of 16 carriers, 2 ends per carrier, 0.0126±0.0005 inch (.320±.013 mm) wire applied with 2.33± 0.5 picks per inch; or 24 carriers, 2 ends per carrier, 0.010±0.0005 inch (.254±.013 mm) wire applied with 3.50±0.50 picks per inch; or 32 carriers, single end per carrier, 0.0126±0.0005 inch (.320±.013 mm) wire applied with 4.67± 0.50 picks per inch. Tensile strength, type, weight and uniformity of coating shall be as specified in 3.1.1.1.1. Loose strands in the braid are not permitted. Joints in steel strands shall be welded or brazed. A brazed or welded joint shall be the same diameter as the original strand. Dressing of joints or removal of burrs is acceptable to meet these requirements. Welded joints shall be annealed to remove any brittleness. Joints in steel strands shall have an average tensile strength of not less than 35 percent of the average tensile strength of the strands adjacent to the joints, when tested on not less than 12 samples. Joints in the steel strands of the braid shall be so staggered so o detrimental effects on the flexing and tensile properties of the cable. Steel strands that break during the braiding operation may be repaired. This may be accomplished by trimming and then soldering the loose ends to wires running parallel to themselves. A lead-tin alloy solder, a noncorrosive flux, and a fine soldering iron may be used. The length of the soldered section shall be not than .250 inch (6.35 mm). The free ends shall be trimmed flush so that no free ends exist. The soldered joint shall be dressed down with a file or other suitable means so that no appreciable increase in braid diameter nor a decrease in jacket thickness develops (see 4.6.10).

3.1.11 Outer jacket. A tightly fitting, seamless jacket of black, low temperature polyvinylchloride jacketing compound such as B.F. Goodrich GEON 8677, Bakelite Compound QFD-9924 or QFD-9975, Union Carbide W919, or equal, shall be applied over the braid. The jacketing compound shall fill the interstices of the braid. The nominal thickness shall be 0.047 inch (1.19 mm) and the minimum thickness shall be not less than 0.042 inch (1.07 mm). The diameter of the finished cable shall be not greater than 0.625 inch (1.586 mm) and not less than 0.600 inch (15.24 mm) (see 4.6.11).

3.1.11.1 Tubing requirement of jacket. To determine whether the extrusion process has caused strains in the jacket, a specimen of the cable shall be subjected to the test specified in 4.6.11.1. The jacket shall show no evidence of cracking when examined under a magnifying glass of approximately 3 diameters magnification (focal distance 8 cm).

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3.1.11.2 Physical requirements of the jacket before aging. The outer jacket removed from the cable or taken from a slab section shall meet the following requirements performed at an ambient temperature of 18 to 32 °C (see 4.6.11.2).

a. Tensile strength. - The jacket specimen shall have a tensile strength of not less than 2,000 pounds per square inch of original cross sectional area at rupture.

b. Elongation. - The jacket specimen shall have an elongation of 350 percent minimum at rupture.

3.1.11.3 Physical requirements of the jacket after aging. The jacket specimen obtained as indicated in 3.1.11.2 shall meet the following physical requirements when tested at a temperature of 18 to 32 °C after the specimen has been subjected to the aging process specified in 4.6.11.3.

a. Tensile strength. The jacket specimen shall have a tensile strength of not less than 90 percent of the value obtained on an adjacent specimen prior to artificial aging.

b. Elongation. The aged jacket specimen shall have an elongation of the 2 inch gauge marks at rupture, not less than 90 percent of the value obtained on an adjacent specimen prior to artificial aging.

3.1.12 Identification marking. The cable shall be marked on the outside with an inked marking. The outer surface of the cable shall be smooth and free from raised or indented markings. All letters and numbers shall be of the same height; inked markings shall be clear and legible. The inked marking shall consist in succession of the nomenclature, the specification number, manufacturer's company initials and year of manufacture +30 days. Three to five letter spaces shall be allowed between them. The marking shall be repeated at intervals along the cable not to exceed 1 foot (see 4.6.12).

3.2 Electrical requirements (see 4.7).

3.2.1 DC resistance. The direct current resistance of each conductor shall not exceed 62Ω per 1000 ft. loop at 20 °C (see 4.7.1).

3.2.2 Dielectric strength. There shall be no evidence of voltage drop when tested in accordance with 4.7.2.

3.2.3 Insulation resistance. When tested in accordance with 4.7.3 the insulation resistance shall be not less than 10,000 megohms per 1,000 feet. of cable at 20 °C.

3.2.4 Conductor resistance unbalance. The conductor resistance unbalance for each pair shall not exceed 4 percent except that not more than 2 pairs per length may have values between 4 and 8%. The conductor resistance unbalance in each pair is expressed as a percentage as follows: (see 4.7.4).

$$\frac{\text{(Higher Value-Lower Value) X 100}}{\text{Lower Value}}$$

3.2.5 Mutual capacitance. In each length of cable, the mutual capacitance of each pair measured at 60 °F at a frequency of 1000±100Hz shall not exceed 0.090 µF per mile, when tested in accordance with 4.7.5.

3.2.6 Capacitance unbalance. In every length of cable, the 1000Hz pair to pair capacitance unbalance shall not exceed 160 micro-microfarads for any length of 1,000 feet or less. The average pair to pair capacitance unbalance shall not exceed 40 pF for any length of 1000 feet or less. For other lengths of

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cable, the values shall not exceed the value obtained by multiplying the value for 1000 feet by the square root of the ratio of the length in question to 1000 feet (see 4.7.6).

3.2.7 Attenuation. The attenuation at 1000Hz at 20 °C shall be not greater than 2.7db per mile of cable terminated with 600Ω of resistance or when computed from measured primary cable characteristics: resistance and capacitance (see 4.7.7).

3.2.8 Spark test. The jacket shall withstand a spark test of 7,500 V rms when tested as specified in 4.7.8.

3.3 Mechanical requirements (see 4.8).

3.3.1 Extreme temperature (aging). Following the tests specified in 4.8.1, there shall be no visible evidence of cracking of the jacket or the steel braid when examined under a magnifying glass of approximately 3 diameters magnification (focal distance 8 cm). The first and last turns shall be discounted.

3.3.2 Breaking strength. The cable shall be prepared as specified in 4.8.2. A 100 pound static load shall be applied initially for a period of one half hour. The cable shall be marked with tape to indicate relationship between cable core, cable jack and cable grip. The static load shall be increased to 800± 20 pounds for a period of 16 hours. Following the test, the cable shall show no rupture of the jacket and slippage of the cable core shall not exceed .500 inch (12.7 mm) (see 4.8.2).

3.3.3 Flexing. After being subjected to 2000 cycles of the flex test of 4.8.3 the cable shall meet the requirements of 3.2.1 dc resistance, 3.2.2 dielectric strength, and 3.2.3 insulation resistance. In addition, there shall be no physical damage to the cable. Upon completion the portion of cable tested shall be discarded.

3.3.4 Twisting. After being subjected to 2000 cycles of the flex test of 4.8.4 the cable shall meet the requirements of 3.2.1 dc resistance, 3.2.2 dielectric strength, and 3.2.3 insulation resistance. In addition, there shall be not physical damage to the cable. Upon completion the portion of cable tested shall be discarded.

3.3.5 Impact. After being subjected to 200 cycles of the impact test of 4.8.5, the cable shall meet the requirements of 3.2.1 dc resistance, 3.2.2 dielectric strength, and 3.2.3 insulation resistance. In addition, there shall be physical damage to the cable. Upon completion the portion of cable tested shall be discarded.

3.4 Repairs. No repair splices shall be made in the finished cable as a whole. Repairs made in the cable components during manufacture shall conform to the following requirements (see 4.9).

3.4.1 Stranded conductor splices. When splices are required in the individual strands, they shall be of the butt type. A brazed or welded joint shall be the same diameter as the original wire. Dressing of splices or removal of burrs is acceptable to meet this requirement. Multiple strand splices may be made in the stranding operation by joining the individual strands provided that the individual strand splices are spaced a minimum of 1 foot apart. No strands shall be kinked. There shall be no splices or joints in the conductor as a whole (see 4.9.1.2).

3.4.1.1 Copper strands. Copper strand splices shall be of the butt type, brazed with silver solder (see 4.9.1.1).

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3.4.1.2 Steel strands. Steel strand splices shall be of the butt type, welded or brazed with silver solder. Joints shall be annealed to remove any brittleness. The repaired strand shall be not less than 35 percent of the average tensile strength of the strand adjacent to the joint (see 4.9.1.2).

3.4.2 Insulation repairs. All insulation repairs shall be made with a suitable heat stabilized, non-oxidizing polyethylene compound and shall be of the heat molded type. In making such repairs, the original insulation shall be removed by smooth tapering cuts, extending on each side of the defect without nicking the conductor. The finished repair shall be cylindrical and smooth with no flash of excess patching compound. The minimum wall thickness requirements of 3.1.2 shall be maintained as closely as possible. The maximum overall length of a repair shall be 4 inches (102 mm). Insulation repairs shall be sufficiently flexible to meet the low temperature requirements of 3.3.1 using a mandrel of 0.160±0.005 inch (4.06±.127 mm). There shall be no evidence of cracking of the patch or of the joint between the patch and the original insulation (see 4.9.2).

3.4.3 Jacket repairs. If repairs are required in the inner or outer jackets, the materials and methods used shall be subject to the approval of the contracting officer (see 4.9.3).

3.5 Workmanship. The cable shall be manufactured in such a manner as to be of uniform quality and free from any defects that would affect life, serviceability or appearance (see 4.9.4).

3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).

b. Conformance inspection (see 4.5).

4.2 Test equipment and inspection facilities. Unless otherwise specified in the contract or order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.3. First article. Unless otherwise specified in the contract or order, the first article inspection shall be performed by the contractor.

4.3.1. First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for group A, group B, and group C (see tables II, IV, and VI respectively). The inspection shall be performed in the following order: 4.4, group A and group B for all units; and group C as specified in the solicitation or contract. After

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completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspections.

4.3.2 First article test plan and report(s). The first article test plan and test report(s) shall be as required in the solicitation or contract.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Requirement paragraph
Copper strands	3.1.1.1
Galvanized steel-strand	3.1.1.1.1b,c
Conductor insulation	3.1.2
Core wrap	3.1.8
Inner jacket	3.1.9

4.5 Conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements.

4.5.1 Group A inspection. Each length of cable (100 percent) of each lot shall be inspected for conformance to all the inspections, and test requirements of tables II and III. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Visual and mechanical	3.5	4.9.4	See table III.
Conductors	3.1.1	4.6.1	See table III.
Color code	3.1.3	4.6.3	See table III.
Pair twist	3.1.4	4.6.4	See table III.
Cabling	3.1.5	4.6.5	See table III.
Marker	3.1.6	4.6.6	See table III.
Binder	3.1.7	4.6.7	See table III.
Braid	3.1.10	4.6.10	See table III.
Outer jacket	3.1.11	4.6.11	See table III.
Identification marking	3.1.12	4.6.12	See table III.
Electrical			
DC resistance	3.2.1	4.7.1	See table III.
Dielectric strength	3.2.2	4.7.2	See table III.
Insulation resistance	3.2.3	4.7.3	See table III.
Conductor resistance unbalance	3.2.4	4.7.4	See table III.
Mutual capacitance	3.2.5	4.7.5	See table III.
Capacitance unbalance	3.2.6	4.7.6	See table III.
Attenuation	3.2.7	4.7.7	See table III.
Spark test	3.2.8	4.7.8	See table III.

TABLE III. Sample sizes for group A inspection.

Lot size	Visual and mechanical	Electrical
2 to 8	All	All
9 to 150	13	13
151 to 280	20	20
281 to 500	29	29
501 to 1,200	34	34
1201 to 3,200	42	42
3201 to 10,000	50	50
10,001 to 35,000	60	60
35,001 to 150,000	74	74
150,001 to 500,000	90	90
500,001 and over	102	102

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4.5.2 Group B inspection. This inspection shall be listed in table IV and shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Group B sampling plan. A sample of parts shall be randomly selected in accordance with table V. If one or more defects are found, the lot shall be rescreeened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table V; if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Physical requirements of jacket before aging	3.1.11.2	4.6.11.2	See table V.

TABLE V. Sample sizes group B inspection.

Lot sizes	Tensile strength	Elongation
2 to 8	All	All
9 to 150	13	13
151 to 280	20	20
281 to 500	29	29
501 to 1,200	34	34
1,201 to 3,200	42	42
3,201 to 10,000	50	50
10,001 to 35,000	60	60
35,001 to 150,000	74	74
150,001 to 500,000	90	90
500,001 and over	102	102

4.5.3 Group C inspection. This inspection shall be listed in table VI and shall be performed on sample units that have been subjected to and met group A and group B inspection. These units shall be taken from the first cable produced and from every million feet produced or every 60 days whichever occurs first. These samples shall be selected without regard to their quality.

TABLE VI. Group C inspection.

Inspection	Requirement	Test method	Number of sample units
Thermal stress cracking	3.1.2.1	4.6.2.1	6
Physical requirement of jacket after aging	3.1.11.3	4.6.11.3	6
Low temperature test after aging	3.3.1	4.8.1	12
Breaking strength	3.3.2	4.8.2	6
Flexing	3.3.3	4.8.3	1
Twisting	3.3.4	4.8.4	1
Impact	3.3.5	4.8.5	1
Repairs	3.4	4.9	N/A

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4.5.3.1. Noncompliance. The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigations and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of produce which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective actions have been taken additional sample units shall be subjected to the group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups a and b inspections may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective (see 6.4).

4.5.4 Rejected lots. Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Methods of inspection.

4.6.1 Conductor. Six samples of each conductor shall be tested in accordance with 4.6.1.1 and 4.6.2 (see 3.1.1).

4.6.1.1 Copper strands. Copper strands shall be tested for compliance with the requirements of 3.1.1.1.

4.6.1.2 Galvanized steel strands. Steel strand shall be tested in accordance with the following for compliance with the requirements of 3.1.1.1.1.

a. Tensile strength. Test the steel strand to determine that the breaking strength is as required in 3.1.1.2a.

b. Weight of coating. Test the steel strand in accordance with ASTM-A90/A90M to determine that the weight of the coating is as required in 3.1.1.2b.

c. Uniformity of coating. Test several specimens from a single sample of the steel strand in accordance with ASTM A239 to determine the number of 15 second immersions before the end point is reached as required in 3.1.1.2c.

4.6.2 Conductor insulation. Conductor insulation shall be tested for compliance with the requirements of 3.1.2.

4.6.2.1 Thermal stress cracking. Specimens shall be looped back and tightly wound back on themselves for 5 close turns and the ends secured. The specimens shall be placed in an oven at a temperature of 100±1 °C for 14 days. Following the test, the specimens shall cool naturally to room temperature and then examined as specified in 3.1.2.1.

4.6.3 Color code. The insulated conductors shall be inspected for compliance with requirements of 3.1.3.

4.6.4 Pair twist. Cable shall be inspected for compliance with the requirements of 3.1.4.

4.6.5 Cabling. Cable shall be inspected for compliance with the requirements of 3.1.5.

4.6.6 Manufacturer's marker. Cable shall be inspected for compliance with the requirements of 3.1.6.

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4.6.7 Binding. Cable shall be inspected for compliance with the requirements of 3.1.7.

4.6.8 Core wrap. Cable shall be inspected for compliance with requirements of 3.1.8.

4.6.9 Inner jacket. Cable shall be inspected for compliance with the requirements of 3.1.9.

4.6.10 Braid. Cable shall be inspected for compliance with the requirements of 3.1.10.

4.6.11 Outer jacket. Cable shall be inspected for compliance with the requirements of 3.1.11.

4.6.11.1 Tubing test of outer jacket. Six specimens of the jacketed cable shall be wound tightly on a mandrel having a diameter of 0.625±0.005 inches (1.59±.13 mm) for 5 close turns. The ends shall be anchored securely and the specimen placed in an oven maintained at a temperature of 120±1 °C for 1 hour. The specimens shall meet the requirements of 3.1.11.1.

4.6.11.2 Physical test of jacket before aging. The jacket shall be tested for compliance with the requirements of 3.1.11.2.

4.6.11.3 Physical test of jacket after aging. Six samples of the jacket shall be tested for compliance with the requirements of 3.1.11.3. Aging shall be performed as follows: Place the specimen in an air oven, conforming to the requirements of ASTM D573, and maintain at a temperature of 80±1 °C for a period of 14 days, and follow with a cooling period of at least 24 hours at room temperature.

4.6.12 Identification marking. Cable shall be inspected for compliance with the requirements of 3.1.12.

4.7 Electrical tests (see 3.2).

4.7.1 Direct current resistance. Each pair of cable length under test shall be tested for compliance with the requirements of 3.2.1.

4.7.2 Dielectric strength. The insulation shall be capable of withstanding for not less than 3 seconds a dc potential of 3.0 kV or an ac potential of 2100 V rms applied between each conductor and the remaining conductors connected to the braid (see 3.2.2).

4.7.3 Insulation resistance. Immediately following the dielectric strength test, the insulation resistance between each conductor in turn and the remaining conductors connected to the braid and grounded shall meet the requirements of 3.2.3. In performing the insulation resistance test, an electromotive force no less than 100 V shall be used, the conductors being maintained negative with respect to the ground. The insulation resistance shall be computed form the galvanometer deflection obtained after an electrification of 1 minute. As an alternate, a direct reading insulation resistance bridge may be used. In making the insulation resistance test, the test may be terminated in less than 1 minute if the galvanometer has ceased fluctuating and the reading indicates the required minimum insulation resistance has been obtained. However, tests results on 5 percent of the lengths after 1 minute electrification shall be recorded to permit a continuous check on quality.

4.7.4 Conductor resistance unbalance. Each cable length shall be tested for compliance with the requirements of 3.2.4. One pair from the center layer, 2 pairs from the 1st layer, and 3 pairs from the 2nd layer shall be selected at random and tested for conductor resistance unbalance.

4.7.5 Mutual capacitance. Each cable length shall be tested for compliance with the requirements of 3.2.5. One pair from the center layer, 2 pairs form the 1st layer, and 3 pairs from the 2nd layer shall be selected at random and tested for mutual capacitance. The capacitance shall be measured at a

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temperature not lower than 60 °F. If the cable, when measured at a temperature higher than 60 °F, fails to meet the requirements, no correction factor may be applied. In such case, the cable may be retested at a temperature no less than 60 °F. The mutual capacitance of a pair is that measured between the 2 wires of a pair, the remainder of the conductors being connected to the braid, which shall be grounded.

4.7.6 Capacitance unbalance. Each cable length shall be tested for compliance with the requirements of 3.2.6. Capacitance unbalance shall be measured between each pair and its adjacent pairs in each layer, and between each pair in a layer and all other pairs in adjacent layers.

4.7.7 Attenuation. Each cable length shall be tested for compliance with the requirements of 3.2.7. One pair from the center layer, 2 pairs from the 1st layer, and 3 pairs from the 2nd layer shall be selected at random and tested for attenuation.

4.7.8 Spark test. The finished cable shall be run through a drag chain spark tester at a speed sufficient to enable any point of the cable to be subjected to the voltage in 3.2.8 for .025 seconds minimum.

4.8 Mechanical tests (see 3.3).

4.8.1 Low temperature test after aging. Twelve samples of cable WM-130( )/U shall be subjected to air oven aging and cooling as outlined in paragraph 4.6.11.1. The samples shall then be subjected to the tests of 4.8.1 and 4.8.1.2 (see 3.3.1).

4.8.1.1 Low temperature requirement at – 40 °C. Six samples of the completed cable and a test mandrel whose diameter shall be 1.688± 0.005 inches (42.88 mm ±.13 mm) shall be placed in a cold chamber maintained at a temperature of –40±1 °C for 24 hours. At the end of this time, and while still in the cold chamber, the cable shall be wound around the mandrel for 7 close turns. No object having a higher temperature than –40±1 °C shall come within 12 inches (305 mm) of the part of the specimen which is under test (see 3.3.1).

4.8.1.2 Low temperature requirement at –55 °C. Six samples of the completed cable and a test mandrel whose diameter shall be 3.75 inches + 0.010 inches shall be placed in a cold chamber maintained at a temperature of –55±1 °C for 24 hours. At the end of this time, and while still in the cold chamber the cable shall be wound around the mandrel for 3 turns at a slow rate. No object having a higher temperature than –55±1 °C shall come within 12 inches (305 mm) of the part of this specimen which is under test (see 3.3.1).

4.8.2 Breaking strength. Three samples of cable shall be tested from every 10,000 feet of cable produced or every ten days, whichever comes first. The sample shall be tested using cable grip shown on CECOM drawing SM-B-159154. The grip shall be secured to a stationary object with a minimum of four feet of cable within the grip. The other end of the cable shall be wound on a mandrel and static load shall be applied between the cable, the cable and the cable grip. The ferrule on the cable grip may be encased in a protective device to prevent collapse of the ferrule when under stress (see 3.3.2).

4.8.3 Flexing. The end of one cable length shall be subjected to the flex test as described on figure 1 using a bending diameter of .625± .0156 inch (15.86±.397 mm). The test shall be conducted at room temperature. The samples shall meet the requirements of 3.3.3.

4.8.4 Twisting. The end of one cable length shall be subjected to the twist test as described on figure 2. The test shall be conducted at room temperature. The sample shall meet the requirements of 3.3.4.

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4.8.5 Impact. The end of one cable length shall be subjected to the impact test as described on figure 3. The test shall be conducted at room temperature. The sample shall meet the requirements of 3.3.5.

4.9 Repairs.

4.9.1 Stranded conductor splices. Conductor strand splices shall conform to the requirements of 3.4.1.

4.9.1.1 Copper strands. Copper strand splices shall conform to the requirements of 3.4.1.1.

4.9.1.2 Steel strands. Test the steel strand joint to determine that the tensile strength of the strand containing the joint is equal to not less than 35 percent of the average tensile strength of the strand adjacent to the joint (see 3.4.1.2).

4.9.2 Insulation repairs. Test specimens of repaired insulation to determine compliance with the requirements of 3.4.2.

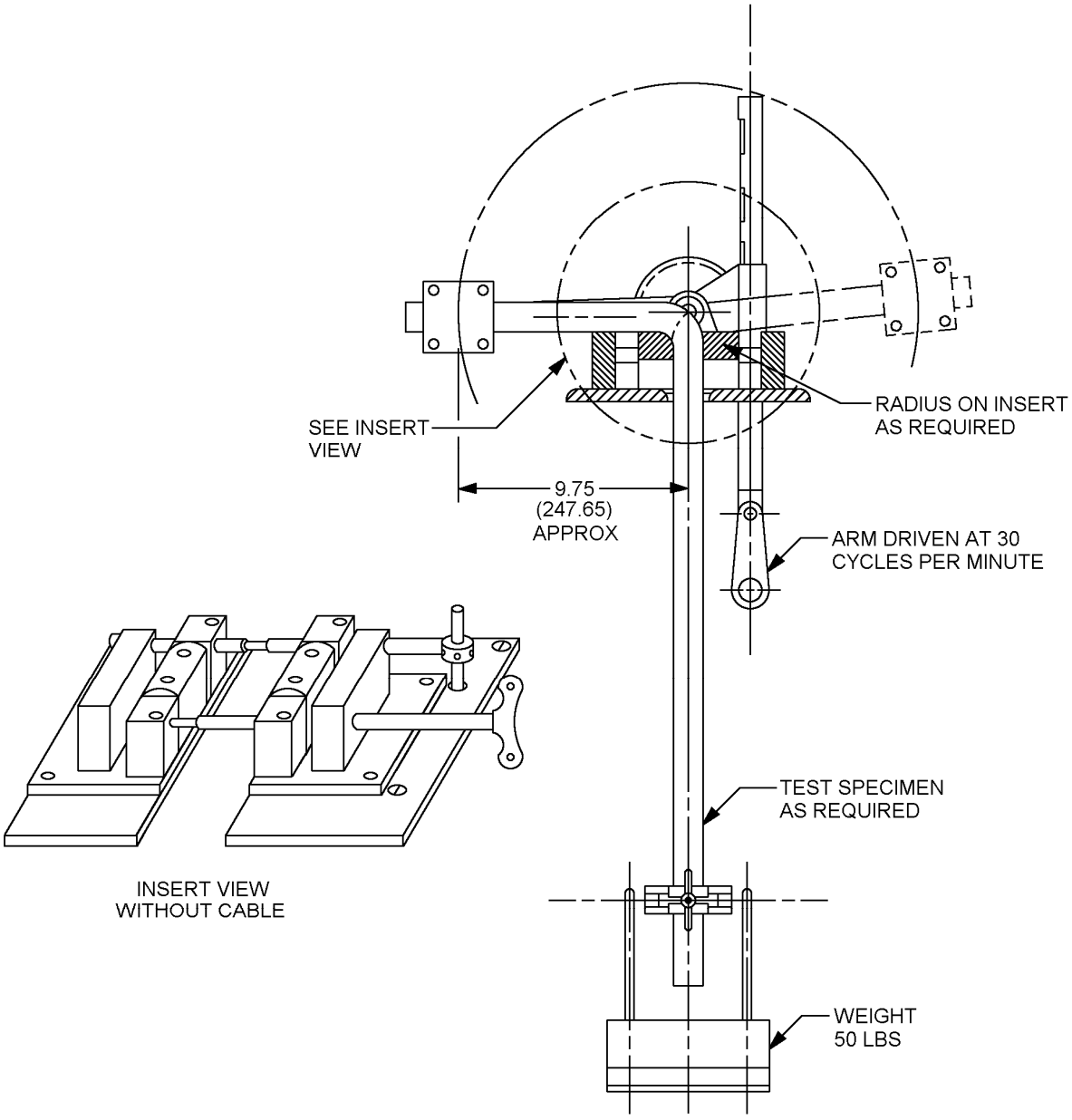
4.9.3 Jacket repairs. Jacket repairs shall be in accordance with the contracting officer's written approval (see 3.4.3).

4.9.4 Visual and mechanical inspection. Equipment shall be examined for the defects listed in table VII and meet the requirements of 3.5.

TABLE VII. Visual and mechanical defects.

Defect
Incorrect number of strands.
Copper strands not correct size.
Steel strands not correct size.
Color code not correct or clear.
Pairing not correct.
Cabling not correct.
Inner jacket not correct thickness.
Braid not correct size or adequate.
Outer jacket not correct thickness.
Cable diameter not correct.
Conductor not centered within insulation.
Core wrap inadequate or not correctly wound.
Inner jacket not centered.
Braid contains loose strands.
Outer jacket not centered.
Identification marking not correct.

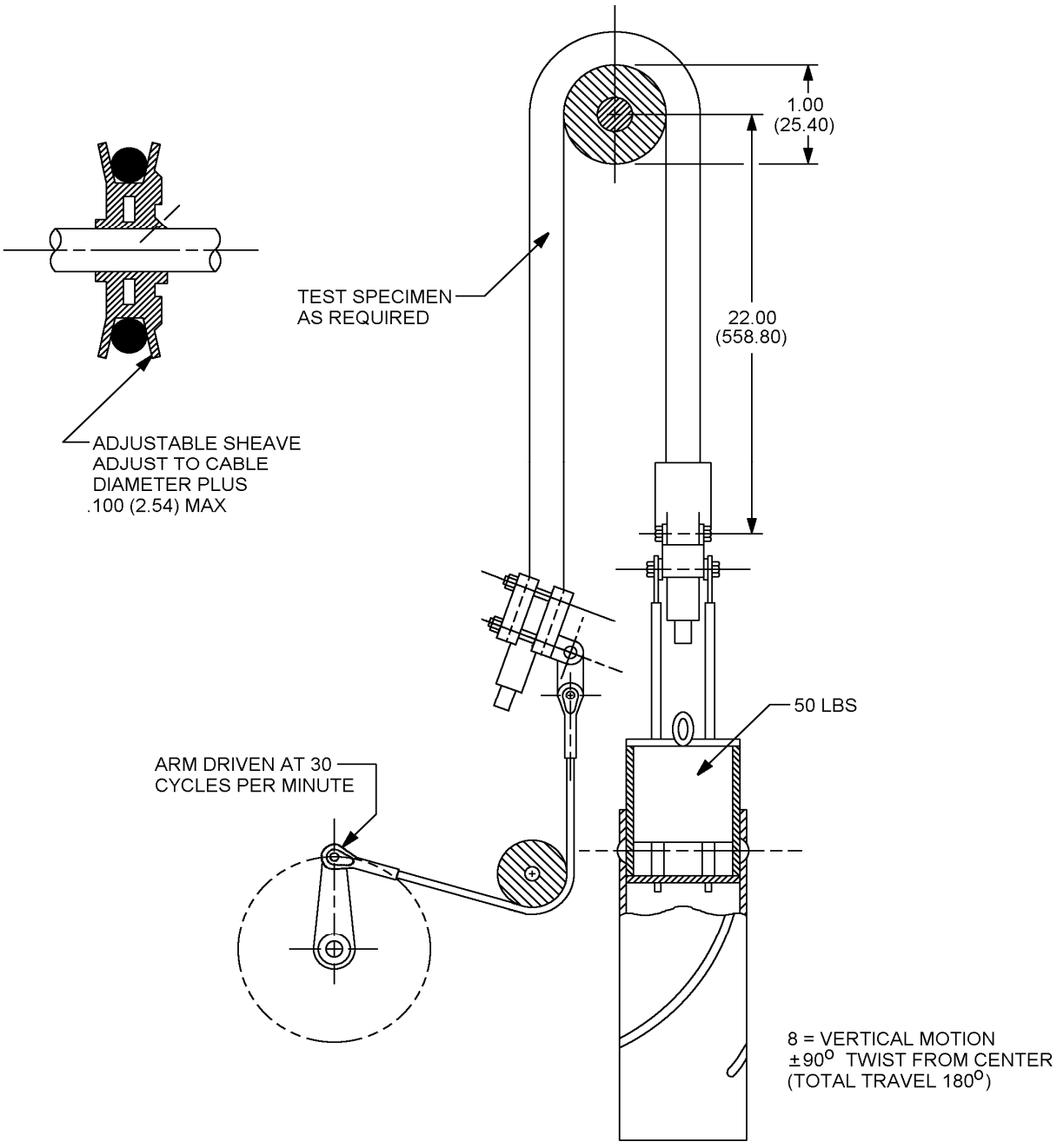
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NOTES:
1. During test a 60 cycle AC current of not less than 0.5A flow through conductors.
2. 1 cycle shall consist of 90º bend each side and return to the original position.

FIGURE 1. Flex test fixture, cable testing

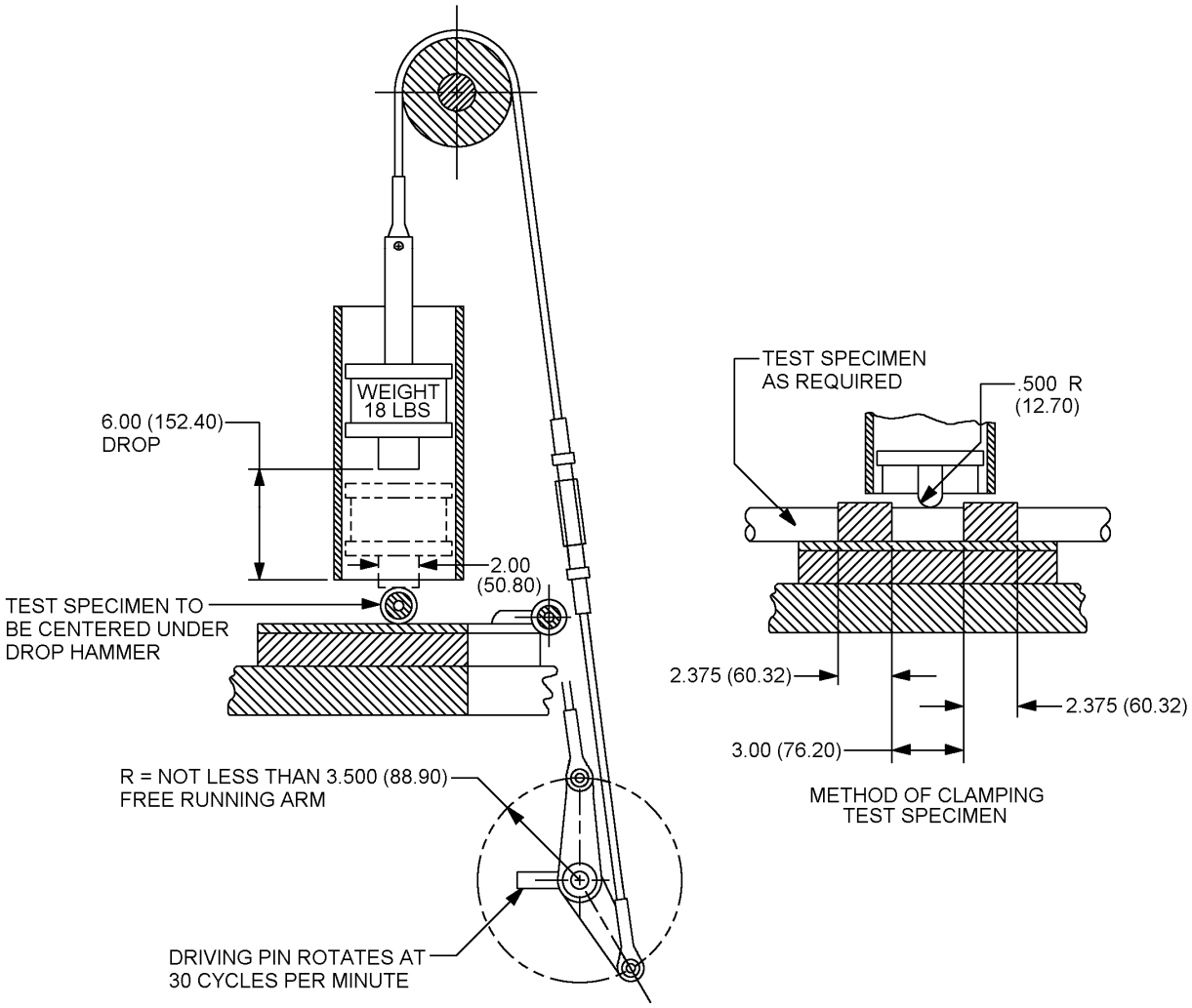
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NOTES:
1. Twist per inch of active length to be kept constant when changing pulley and larger clamping arm.
2. During test a 60 cycle AC current of not less than 0.5A to flow through conductors.

FIGURE 2. Twist test fixture, cable testing

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NOTE: During test, a 60 cycle AC current of not less than 0.5A to flow through conductors.

FIGURE 3. Impact test fixture, cable testing

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's System Commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature, which may be helpful, but is not mandatory).

6.1 Intended use. The cable is intended for use in field telephone voice frequency applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. The specific issue of individual documents referenced.

c. Complete type designation.

d. Packaging requirements.

6.3 Definitions.

6.3.1 Unit of product. The unit of product is defined as a length of cable.

6.3.2 Length. The length is defined in the invitation for bid and contract.

6.3.3 Lot size. The lot size as determined to be the number of units of products as determined in 6.3.1 offered for inspection.

6.3.4 Sample. The sample consists of that number of sample units required by the sampling plan for the lot size determined as in 6.3.3.

6.3.5 Selection of sample. The number of sample units required for inspection chosen by selecting 1 coil, spool, or reel for each sample unit required, after which the coil, spool or reel will be treated as the sample unit for purposes of inspection.

6.3.6 Specimen. A specimen is an individual piece of cable taken from a sample unit.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.1).

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6.5 Subject term (keyword) listing.

Conductor, coated
Conductor, stranded
Galvanized steel wire
High density
Non-rigid
Polyethylene
Polyvinylchloride

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.7 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example, WM-130 W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.9 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. Table VIII lists the Environmental Protection Agency (EPA) top seventeen hazardous materials targeted for major usage reduction. Use of these materials should be minimized or eliminated unless needed to requirements specified herein (see section 3).

TABLE VIII. EPA top seventeen hazardous materials.

Benzene	Dichloromethane	Tetrachloroethylene
Cadmium and Compounds	Lead and Compounds	Toulene
Carbon Tetrachloride	Mercury and Compounds	1,1,1 Trichoroethane
Chloroform	Methyl Ethyl Ketone	Trichloroethylene
Chromium and Compounds	Methyl Isobutyl Ketone	Xylene
Cyanide and Compounds	Nickel and Compounds	

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CONCLUDING MATERIAL

Custodians:
Army – CR
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2005-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, please verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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